Jashore University of Science and Technology

Bachelor of Science Electrical and Electronic Engineering

1st semester of 1st year (2025–2026)

Course no.: 053311 PHY 1101 **Course title: Physics**

Class test no.: 02 **Date: 05 September 2026**

Roll:	

1. Does increasing the intensity of incident light cause photoelectric emission when its frequency is below the threshold frequency? [1]

Yes No

2. Is the Compton wavelength shift dependent on the wavelength of the incident photon? [1]

Yes No

3. Did Bohr postulate that electrons can exist in certain stable stationary orbits without continuously radiating energy? [1]

Yes No

4. Does the wave function ψ itself represent a directly measurable physical quantity? [1]

Yes No

5. Can the position and momentum of a particle both be measured with unlimited precision simultaneously? [1]

Yes No

6. A metal has a work function of 2.2 eV. When light of wavelength 300 nm falls on it, determine the stopping potential of the emitted photoelectrons. [2.5]

7. An X-ray photon of wavelength 0.071 nm is scattered from a free electron, and its wavelength after scattering is 0.07343 nm. Calculate the scattering angle. [2.5]